DATA ANALYTICS PROJECT

Report

On

Data Visualisation Story Line

Computer Science & Engineering (CSE)

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Submitted to:

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected faculty guide, **Mr. Vishal Dutt**, for his invaluable guidance, unwavering support, and constant encouragement throughout the development of this project. His clear explanations and constructive feedback helped us understand data visualization concepts deeply and complete the project in a well-organized manner. Without his direction and supervision, this work would not have been possible.

We are also thankful to the Department of Computer Science Engineering at Chandigarh University for providing us with the necessary academic resources, labs, and a positive learning environment. We extend our appreciation to all teaching and non-teaching staff members who supported us during different stages of the project.
We are grateful to our team members — Ronak Lodhi, Archit Sharma, and Jatin Kumar — for their dedication, cooperation, and teamwork. Each member contributed their best efforts and worked responsibly to achieve our common goal. Finally, we express our heartfelt gratitude to our parents and well-wishers for their constant motivation and encouragement throughout our academic journey. Their support has always been our greatest strength.

ABSTRACT

In the modern data-driven world, raw numbers and spreadsheets rarely communicate insights on their own. This project focuses on **transforming raw data into meaningful visual stories** using Python. The main objective is to analyze datasets and present insights in a structured and logical storytelling format. The system uses data cleaning, analysis, and visualization techniques to convert numerical data into understandable graphical outputs.

The dataset is first cleaned and preprocessed by handling missing values, removing duplicate records, and correcting inconsistencies. After data preparation, descriptive statistical methods such as mean, median, and standard deviation are applied. Various visualization techniques — including bar charts, line graphs, pie charts, scatter plots, and heatmaps — are used to reveal patterns and relationships.

A key innovation of this project is the **storyline approach**: each visualization is accompanied by a logical narrative explaining what the chart shows and why it matters. This bridges the gap between data analysis and decision-making. The final outcome helps users — including business analysts, managers, and students — interpret trends, patterns, and insights effectively without requiring deep technical knowledge.

INTRODUCTION

In today's digital world, organizations generate enormous volumes of data every day through sales, customer purchases, marketing activities, and operational processes. This data contains valuable information, but raw data alone is not useful unless it is properly analyzed and communicated. A key challenge is not just analyzing data, but presenting it in a way that non-technical stakeholders can understand and act upon.

Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data. Python libraries such as **Pandas, Matplotlib, and Seaborn** have made it possible to automate and enhance this process significantly.

This project introduces a structured **Data Visualisation Story Line** framework that goes beyond simply plotting graphs. It follows a complete narrative sequence: dataset introduction → data cleaning → exploratory analysis → key insights → conclusion. Each graph is supported with interpretation explaining what the data shows and why it is important. This makes visualization more meaningful, analytical, and decision-ready.

LITERATURE REVIEW

Data visualization is the graphical representation of information using charts, graphs, and dashboards. Modern research emphasizes the importance of storytelling in visualization, where data is not only presented visually but also explained logically. Python libraries such as Pandas, Matplotlib, and Seaborn are widely used for analyzing and visualizing datasets. Story-driven visualization improves understanding, engagement, and decision-making.

Edward Tufte (1983) pioneered the field of data visualization and introduced the concept of 'data-ink ratio', arguing that every element in a chart should serve a purpose. His principles continue to guide modern visualization design. Cole Nussbaumer Knaflic (2015) extended this thinking with narrative-driven data presentation, demonstrating that context and storytelling are as important as the data itself.

Research by Few (2009) shows that visual representation of data allows decision-makers to quickly understand information and take action. Tools like Python (Matplotlib, Seaborn), Excel, Power BI, and Tableau are commonly used. However, most existing tools focus on chart generation without providing structured narrative interpretation — a gap that this project directly addresses.

Several studies confirm that users retain information more effectively when data is presented with a logical storyline rather than isolated charts. This project builds upon these findings by implementing a systematic storytelling methodology across the full data analysis pipeline.

EXISTING WORK

Traditional visualization tools generate static charts without proper narrative explanation. Most systems display graphs but do not interpret insights clearly. A review of existing solutions reveals several important limitations:

- **Excel Dashboards** — Offer basic charting but lack structured storytelling and logical flow between visualizations.
- **Tableau & Power BI** — Powerful but require significant setup, licensing costs, and technical expertise to configure meaningful narratives.
- **Jupyter Notebooks** — Provide code + charts together, but narrative quality depends entirely on the individual author with no enforced structure.
- **Auto-EDA Tools (Pandas Profiling, Sweetviz)** — Generate automated summaries but lack business-context interpretation and storytelling flow.

These gaps highlight the need for a system that combines automated analysis with structured, human-readable narrative interpretation — which is the core contribution of this project.

PROPOSED SYSTEM

Our system introduces a **structured storytelling approach** in data visualization. It follows a sequential pipeline: dataset introduction, data cleaning, exploratory analysis, key insights, and conclusion. Each graph is supported with interpretation explaining what the data shows and why it is important. This makes the visualization more meaningful and analytical.

Key features of the proposed system include:

- ✓ Automated data ingestion from CSV and Excel files
- ✓ Intelligent data cleaning with detailed change logs
- ✓ Multi-chart visualization suite (bar, line, pie, scatter, heatmap)
- ✓ Narrative text generation for each visualization
- ✓ Structured report output in PDF and HTML formats
- ✓ Role-based insight delivery (analyst vs. manager views)

METHODOLOGY

1. Data Collection

Importing the dataset from CSV or Excel files using Pandas. The dataset is validated for completeness and basic structure before any processing begins.

2. Data Cleaning

Removing null values and duplicate records. Standardizing data types, correcting inconsistent entries, and logging all changes made during the cleaning process.

3. Data Analysis

Applying statistical analysis including measures of central tendency, dispersion, and correlation. Identifying trends, seasonality, and outliers in the dataset.

4. Data Visualization

Creating graphs including bar charts, line graphs, pie charts, and scatter plots using Matplotlib and Seaborn. Each chart is formatted for clarity and professional presentation.

5. Storyline Creation

Writing structured narrative explanations for each visualization. Explaining what the chart shows, what patterns are visible, and what business actions the insights recommend.

6. Report Generation

Presenting all findings in a structured format — combining charts and narrative into a cohesive, professional report suitable for both technical and non-technical audiences.

DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram represents how data moves through the system from input to final visual story output.

Level 0 DFD — System Overview

At the highest level, the system takes a raw Dataset and Business Questions as input from the Business Analyst / User, processes them through the Data Visualisation Story Line System, and delivers Insights & Visual Reports as output. Optionally, cleaned and enriched data may be stored back for future use.

Level 1 DFD — Internal Processes

Process	Description
P1: Data Ingestion	Raw dataset loaded from CSV/Excel into the system pipeline
P2: Data Cleaning	Null values removed, duplicates dropped, formats standardized
P3: Analysis Engine	Statistical computation, trend detection, correlation analysis
P4: Visualization Module	Charts generated with Matplotlib/Seaborn for each insight
P5: Narrative Generator	Textual story created for each chart explaining findings
P6: Report Builder	Final PDF/HTML report compiled and delivered to the user

ROLES AND RESPONSIBILITIES

Ronak Lodhi

- Data Collection and Cleaning
- Dataset Preparation and Validation
- Preprocessing Implementation in Python
- Documentation of data pipeline steps

Archit Sharma

- Data Analysis and Statistical Computation
- Identifying Patterns, Trends, and Correlations
- Exploratory Data Analysis (EDA) implementation
- Chart selection and design decisions

Jatin Kumar

- Data Visualization using Matplotlib and Seaborn
- Designing the Storyline Flow and Narrative
- Report Documentation and Formatting
- Final presentation and output generation

CONCLUSION

The **Data Visualisation Story Line** project successfully demonstrates how raw data can be converted into meaningful, narrative-driven insights using Python. By combining automated analytics with structured storytelling, the system bridges the critical gap between data analysis and business communication.

This project proves that hidden inside every dataset is a goldmine of insights, waiting for someone to dig them out and present them clearly. We started with nothing but CSV files full of numbers — rows and rows of transactions, products, and dates. Through the process of exploratory analysis and visualization, we breathed life into those numbers and transformed them into a clear, logical, and compelling story about the data.

The structured approach — from data ingestion through cleaning, analysis, visualization, and narrative generation — ensures logical flow and better comprehension. The project improves clarity, engagement, and decision-making capability for both technical and non-technical audiences.

Looking ahead, this framework can be extended to include machine learning-based predictive narratives, real-time data feeds, and interactive dashboards. The foundation laid by this project creates readiness for future advanced analytical applications in business intelligence and data journalism.